

SECTION – A

GENERAL APTITUDE

Choose the most appropriate option.

(Q.No. 01 to 13)

1. A Professor passed one sixth of his life in childhood, one twelfth in youth, and one seventh more as a bachelor; five years after his marriage a son was born who died four years before his father at half his final age then what is the age of professor?

(A) 84 years

(B) 74 years

(C) 64 years

(D) 54 years

2. Which word does NOT belong with the others?

(A) wing

(B) fin

(C) beak

(D) rudder

3. Look at this series : 5000, 1001, 201, 41,... What number should come next?

(A) 9

(B) 10

(C) 11

(D) 42

4. Candid is to indirect as honest is to :

(A) frank

(B) wicked

(C) truthful

(D) untruthful

5. Let us consider the length of the side of a square represented by $2y + 3$. The length of the side of an equilateral triangle is $4y$. If the square and the equilateral triangle have equal perimeter, then what is the value of y ?

(A) 3

(B) 4

(C) 6

(D) 8

6. If finch related to bird i.e. FINCH: BIRD. Then, find the pair that has a similar relationship.

(A) frog : toad

(B) elephant : reptile

(C) dalmatian: dog

(D) collie : marsupial

7. Look at this series : 25, 25, 37, 37, __, 51, ... What number should fill the blank?

(A) 51

(B) 39

(C) 23

(D) 25

8. Suppose a fraud shopkeeper sells rice to the customer at the cost price, but he uses a false weight of 900 gm for a kg then his percentage gain is _____.

(A) 5.75 %

(B) 5.56 %

(C) 5.20 %

(D) 5.00 %

9. Divide 88 into four parts such as first part known as a, second part b, third part c, and fourth part is d, when 5 is added to the first part, 5 is subtracted from the second part, 3 is multiplied by

the third part and the fourth part is divided by 5, then all results are equal. Find the value of a, b, c and d respectively.

- (A) 7, 17, 4, 60
- (B) 8, 25, 5, 50
- (C) 10, 30, 3, 45
- (D) 17, 7, 4, 60

10. A software engineer has the capability of thinking 200 lines of code in five minutes and can type 200 lines of code in 10 minutes. He takes a break for five minutes after every ten minutes. How many lines of code will he complete typing after an hour?

- (A) 200
- (B) 300
- (C) 400
- (D) 500

11. A construction company ready to finish a construction work in 180 days, hired 80 workers each working 8 hours daily. After 90 days, only $\frac{2}{7}$ of the work was completed. How many workers are to be increased to complete the work on time?

Note : If additional acquired workers do agree to work for 10 hours daily.

- (A) 90 workers
- (B) 80 workers
- (C) 65 workers
- (D) 85 workers

12. Look at this sequence SCD TEF UGH _____ WKL and find the missing sequence.

- (A) CMN
- (B) UJI
- (C) VIJ
- (D) IJT

13. Here are some words translated from any Language.

diano means oak tree

blynnot means oak leaf

blycrin means maple leaf

Which word could mean “maple syrup”?

(A) blymuth

(B) hupponot

(C) patricrin

(D) crinweel

SECTION – B

ENGINEERING MATHEMATICS

Choose the most appropriate option.

(Q.No. 14 to 30)

14. If $u = f(y - z, z - x, x - y)$, then $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z}$ is equal to:

(A) $x + y + z$

(B) $1 + x + y + z$

(C) 1

(D) 0

15. The general solution of the partial differential equation $(D^2 - D'^2 - 2D')Z = 0$ where

$D = \frac{\partial}{\partial x}$ and $D' = \frac{\partial}{\partial y}$:

(A) $f(y + x) + e^{2x} g(y - x)$

(B) $e^{2x} f(y + x) + g(y - x)$

(C) $e^{-2x} f(y + x) + g(y - x)$

(D) $f(y + x) + e^{-2x} g(y - x)$

16. If y^a is an integrating factor of the differential equation $2xydx - (3x^2 - y^2)dy = 0$, then the value of a is :

(A) -4

(B) 4

(C) -1

(D) 1

17. Using Green's theorem in plane, evaluate $\int_c (2x - y) dx + (x + y) dy$, where c is the circle $x^2 + y^2 = 4$ in the plane:

(A) 2π

(B) 4π

(C) -4π

(D) 8π

18. If $P(A \cap B) = \frac{1}{2}$, $P(\bar{A} \cap \bar{B}) = \frac{1}{2}$ and $2P(A) = P(B) = p$, then the value of p is given by:

(A) $\frac{1}{4}$

(B) $\frac{1}{2}$

(C) $\frac{1}{3}$

(D) $\frac{2}{3}$

19. If $2a + 3b + c = 0$, then at least one root of the equation $ax^2 + bx + c = 0$, lies in the interval:

(A) (0, 1)

(B) (1, 2)

(C) (2, 3)

(D) (1, 3)

20. The following vectors (1, 9, 9, 8), (2, 0, 0, 8), (2, 0, 0, 3) are:

- (A) Linearly dependent
- (B) Linearly independent
- (C) Constant
- (D) None of these

21. While solving the differential equation $\frac{\partial^2 y}{\partial x^2} + 4y = \tan 2x$ by the method of variation of parameters, then value of Wronskion (W) is;

- (A) 1
- (B) 2
- (C) 3
- (D) 4

22. The value of p such that the vector $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ is an eigen vector of the matrix $\begin{bmatrix} 4 & 1 & 2 \\ p & 2 & 1 \\ 14 & -4 & 10 \end{bmatrix}$ is:

- (A) 15
- (B) 16
- (C) 17
- (D) 18

23. A continuous random variable x is distributed over the interval [0, 2] with probability density function $f(x) = ax^2 + bx$, where a and b are constants. If the mean of the distribution is $\frac{1}{2}$. Find the values of the constants a and b.

- (A) $a = 2, b = -\frac{13}{6}$
- (B) $a = -\frac{15}{8}, b = 3$
- (C) $a = -\frac{29}{6}, b = 2$
- (D) $a = 3, b = -\frac{7}{2}$

24. The general solution of the differential equation $\frac{dy}{dx} = (1+y^2)(e^{-x^2} - 2x \tan^{-1} y)$ is:

(A) $e^{x^2} \tan^{-1} y = x + c$

(B) $e^{-x^2} \tan y = x + c$

(C) $e^x \tan y = x^2 + c$

(D) $e^{-x} \tan^{-1} y = x^3 + c$

25. If C is a non-singular matrix and $B = C \begin{bmatrix} 0 & x & y \\ 0 & 0 & z \\ 0 & 0 & 0 \end{bmatrix} C^{-1}$ then :

(A) $B^2 = I$

(B) $B^2 = \text{Null Matrix}$

(C) $B^3 = I$

(D) $B^3 = \text{Null Matrix}$

26. If $w = f(z) = u(x, y) + i v(x, y)$ is an analytic function, then $\frac{dw}{dz}$ is:

(A) $\frac{\partial u}{\partial x} - i \frac{\partial u}{\partial y}$

(B) $\frac{\partial u}{\partial x} + i \frac{\partial v}{\partial y}$

(C) $\frac{\partial u}{\partial x} - i \frac{\partial v}{\partial x}$

(D) $\frac{\partial u}{\partial x} + i \frac{\partial u}{\partial y}$

27. The value of $\int_1^2 \int_0^{x^2} x \, dy \, dx$ is

(A) $\frac{15}{4}$

(B) $\frac{5}{4}$

(C) $\frac{3}{4}$

(D) $\frac{2}{3}$

28. If $f(x) = k \exp. \{ -(9x^2 - 12x + 13) \}$, is a p.d.f. of a normal distribution (k , being a constant), the mean and standard deviation of the distribution:

(A) $\mu = \frac{2}{3}, \sigma = \frac{1}{\sqrt[3]{2}}$

(B) $\mu = 2, \sigma = \frac{1}{\sqrt{2}}$

(C) $\mu = \frac{1}{3}, \sigma = \frac{1}{\sqrt[3]{2}}$

(D) $\mu = \frac{2}{3}, \sigma = \frac{1}{\sqrt{3}}$

29. The value of $\int_C \frac{2z^2 - 5}{(z+2)^2 (z^2+4)z} dz$, (where c is the square with vertices $1 + i, 2 + i, 2 + 2i, i + 2i$) is:

(A) 0

(B) πi

(C) $2\pi i$

(D) $4\pi i$

30. For the function $f(z) = \frac{1}{z^2(e^z - 1)}$, $z = 0$ is a pole of order:

(A) 1

(B) 2

(C) 3

(D) None of these

SECTION – C

COMPUTER SCIENCE

Choose the most appropriate option.

(Q.No. 31 to 85)

31. For the given recurrence equation

$$T(n) = 2T(n-1), \text{ if } n > 0. \\ = 1, \text{ otherwise}$$

- (A) $O(n \log n)$
- (B) $O(n^2)$
- (C) $O(2^n)$
- (D) $O(n)$

32. ____ number of leaf nodes in a rooted tree of n nodes, where each node is having 0 or 3 children.

- (A) $n/2$
- (B) $(2n+1)/3$
- (C) $(n-1)/n$
- (D) $(n-1)$

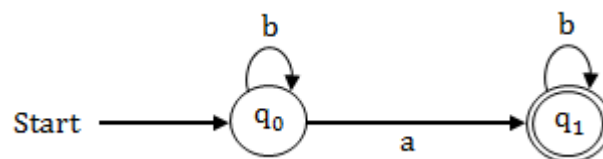
33. ____ number of gates are required to implement the boolean function $(AB+C)$ with using only 2 input NOR gates.

- (A) 2
- (B) 3
- (C) 4
- (D) 5

34. Time To Live(TTL) field in the IP datagram is used ____

- (A) To optimize throughput
- (B) To prevent packet looping
- (C) To reduce delays
- (D) To prioritize packets

35. Consider the following finite state automaton. The language accepted by this automaton is given by the regular.



- (A) ab^*b^*

- (B) $a*b*$
- (C) $b*b$
- (D) $b*ab*$

36. Identify the subnet mask for the given direct broadcast address of subnet of subnet is 201.15.16.31.

- (A) 255.255.192.192
- (B) 255.255.255.198
- (C) 255.255.255.240
- (D) 255.255.257.240

37. The smallest element that can be found in time ____ in a binary max heap

- (A) $O(n \log n)$
- (B) $O(\log n)$
- (C) $O(n)$
- (D) $O(n^2)$

38. The ALU uses ____ to store intermediate result.

- (A) Cache
- (B) Registers
- (C) Accumulators
- (D) Stack

39. ____ symbol is used to denote derived attribute in ER model

- (A) Dashed ellipse
- (B) Square ellipse
- (C) Ellipse with attribute name undirected
- (D) Rectangular box

40. Among all given option, ____ must reside in the main memory.

- (A) Assembler
- (B) Compiler
- (C) Linker
- (D) Loader

41. If the period of a signal is 100 ms. Then its frequency in Hertz is ____

- (A) 10
- (B) 100
- (C) 1000
- (D) 10000

42. $(A+C')(B'+C')$ simplifies to _____

- (A) $AC' + B'$
- (B) $C(A'+B')$
- (C) $BC' + A$
- (D) $AB' + C'$

43. Identify the true statement from the given statements. Program relocation at run time:
1. Requires transfer complete block to some memory locations
 2. Requires both base address and relative address
 3. Requires only absolute address
- (A) (1)
(B) (1) and (2)
(C) (1),(2) and (3)
(D) (1) and (3)
44. ___ is the worst case time complexity for all operations(i.e. Search,update and delete) in a general binary search tree
- (A) $O(n)$
(B) $O(n \log n)$
(C) $O(\log n)$ for search and insert, and $O(n)$ for delete
(D) None of these
45. Maximum number of superkeys for the relation schema $R(X,Y,Z,W)$ with X as the key is:
- (A) 6
(B) 8
(C) 9
(D) 12
46. Identify the correct nodes and edges in the given intermediate code:
- (1) $i=1$
 - (2) $t1=5*i$
 - (3) $t2=4*t1$
 - (4) $t3=t2$
 - (5) $a[t3]=0$
 - (6) $i=i+1$;
 - (7) if $i < 15$ goto(2)
- (A) 33
(B) 44
(C) 43
(D) 34
47. ___ number of queues are needed to implement symbol and is acting as permanent database
- (A) Variable Table
(B) Terminal Table
(C) Keyword Table
(D) Identifier Table
48. Identify the true statement from the given statements.
- (1) Lossless, dependency preserving decomposition into 3NF is always possible

- (2) Any relation with two attributes is in BCNF
 (A) (1)
 (B) (2)
 (C) (1) and (2)
 (D) None of these
49. The hexadecimal representation of $(632)_8$ is:
 (A) 19A
 (B) 198
 (C) 29A
 (D) 291
50. For a given hash table T with 10 slots that stores 1000 elements, the load factor α for T is___
 (A) 100
 (B) 0.01
 (C) 200
 (D) 1.05
51. Identify the true statements from the given statements.
 (1) HTTP may use different TCP connection for different objects of a webpage if non persistent connections are used
 (2) FTP uses two TCP connections, one for data and another control
 (3) TELNET and FTP can only use TWO connection at a time.
 (A) (1)
 (B) (1) and (2)
 (C) (2) and (3)
 (D) (1),(2) and (3)
52. Which of the following languages over the alphabet $\{0,1\}$ is described by the given regular expression: $(0+1)^*1(0+1)^*1$?
 (A) The set of all strings containing the substrings 11
 (B) The set of all string containing at most two 1's
 (C) The set of all strings containing at least two 1's
 (D) The set of all strings that begins and ends with only 0.
53. _____ number of undirected graphs can be constructed using $V=(v_1,v_2,..v_n)$.
 (A) n^3
 (B) $2^{n(n-1)/2}$
 (C) $n^{-1/2}$
 (D) $2^{(n-1)/2}$
54. Dijkstra's algorithm is based on:
 (A) Greedy approach
 (B) Dynamic programming
 (C) Backtracking paradigm
 (D) Divide and conquer paradigm

55. The language $\{W^a X^b Y^{a+b} \mid a, b \geq 1\}$ is:

- (A) Regular
- (B) Context free but non regular
- (C) Context sensitive but non context free
- (D) Type 0 but non context sensitive

56. The process executes the following code and after execution ____ number of child process get created.

```
fork();
```

```
fork();
```

```
fork();
```

```
fork();
```

- (A) 4
- (B) 1
- (C) 15
- (D) 16

57. ____ is used to convert from recursive to iterative implementation of an algorithm

- (A) Array
- (B) Tree
- (C) Stack
- (D) Queue

58. In the truth table, $f(x,y)$ represent the boolean function

x	y	$f(x,y)$
0	0	1
0	1	0
1	0	0
1	1	1

- (A) $x \leftrightarrow y$
- (B) $x \wedge y$
- (C) $x \vee y$
- (D) $x \rightarrow y$

59. Evaluation of the given postfix expression

10 10 +60 6 / * 8 - is:

- (A) 192
- (B) 190
- (C) 110
- (D) 92

60. Identify the total number of tokens in the given statements `printf("A%B=", &i);`

- (A) 7
- (B) 8
- (C) 9
- (D) 13

61. _____ transport layer protocol is used to support electronic mail

- (A) SNMP
- (B) IP
- (C) SMTP
- (D) TCP

62. Find the effective access time for the memory for given data.

Page fault service time=8ms

Average memory access time=20ms

One page fault is generated for every memory access= 10^6

- (A) 29ns
- (B) 33ns
- (C) 28ns
- (D) 30ns

63. Identify the true statement from the given statements

- (1) FIFO is non preemptive
 - (2) Round robin is non preemptive
 - (3) Multilevel queue scheduling is non preemptive
- (A) (1)
 - (B) (1) and (2)
 - (C) (1),(2) and (3)
 - (D) (2)

64. Minimum ____ full adders and half adders are required by the BCD adder to add two decimal digits.

- (A) 3,2
- (B) 9,4
- (C) 6,5
- (D) 5,2

65. Which of the following code replacements is an example of operator strength reduction?

- (A) Replace P^2 by $P * P$
- (B) Replace $P * 16$ by $P << 4$
- (C) Replace $\text{pow}(P, 3)$ by $P * P * P$
- (D) Replace $(P << 5) - P$ by $P * 3$

66. Identify the true statement from the given statements

(1) Number of child pointers in a B/ B+ tree node is always equal to number of keys in it plus one.

(2) B/B+ tree is defined by a term minimum depends on hard disk block size, key and address sizes.

(A) (1)

(B) (1) and (2)

(C) (2)

(D) None of these

67. The following table has two attributes X and Y where X is the primary key and Y is the foreign key referencing X with on delete cascade

X	Y
2	4
3	4
4	3
5	2
7	2
9	5
6	4

The set of all tuples that must be additionally deleted to preserve referential integrity when the tuple (3,4) is deleted is:

(A) (4,3) and (6,4)

(B) (2,4) and (7,2)

(C) (3,2) and (9,5)

(D) (3,4), (4,3) and (6,4)

68. For the given nodes:

89,19,50,17,12,15,2,5,7,11,6,9,100

Minimum _____ number of interchanges are needed to convert it into a max-heap

(A) 3

(B) 4

(C) 5

(D) 6

69. _____ IP address can be used in WAN

(A) 256.0.0.1

(B) 172.16.0.10

(C) 15.1.5.6

(D) 127.256.0.1

70. ____ pairs of traversals is not sufficient to build tree
- (A) Preorder and Inorder
 - (B) Postorder and Inorder
 - (C) Postorder and Preorder
 - (D) None of these
71. Identify the true statement from the given statements:
- (1) A recursive formal language is a recursive subset in the set of all possible words over the alphabet of the language
 - (2) The complement of a recursive language is recursive
 - (3) The complement of a context free language is context free
- (A) Only (1)
 - (B) (1) and (2)
 - (C) (1),(2) and (3)
 - (D) (2) and (3)
72. In the disk, swap space is used to ____
- (A) Save XML files
 - (B) Save process data
 - (C) Save drivers
 - (D) Save HTML files
73. In a system, counting semaphore was initialized to 10. Then 6P(wait) operations and 4V(signal) operations were completed on this semaphore. So ____ is the final value of the semaphore.
- (A) 7
 - (B) 8
 - (C) 13
 - (D) 12
74. _____ to evaluate an expression an execution without any embedded function calls
- (A) Two stacks are required
 - (B) One stack is needed
 - (C) Three stacks are required
 - (D) More than three stacks are required
75. The context free grammar $S \rightarrow aSb \mid bSa \mid \epsilon$ generates:
- (A) Equal number of a's and b's
 - (B) Unequal number of a's and b's
 - (C) Any number of a's followed by any number of b's
 - (D) Any number of b's followed by any number of a's

76. A RAM chip has a capacity of 1024 words of 8 bits each (1K x 8). The number of 2 x 4 decoders with enable line needed to construct a 32K x8 RAM from 1K x8 RAM is:

- (A) 4
- (B) 5
- (C) 6
- (D) 7

77. ____ merges the bodies of two loops.

- (A) Loop rolling
- (B) Loop folding
- (C) Loop merge
- (D) Loop jamming

78. Given message M=1010001101. The CRC for this given message using the divisor polynomial $x^5 + x^4 + x^2 + 1$ is ____

- (A) 01011
- (B) 10101
- (C) 01110
- (D) 10110

79. ____ sorting algorithms has the lowest worst case complexity

- (A) Selection sort
- (B) Bubble sort
- (C) Merge sort
- (D) Quick sort

80. ____ is NOT a part of the ACID properties.

- (A) Inconsistency
- (B) Consistency
- (C) Atomicity
- (D) Isolation

81. ____ number of queues are needed to implement a stack

- (A) 1
- (B) 2
- (C) 3
- (D) 4

82. Baud rate measures the number ____ transmitted per second.

- (A) Symbols
- (B) Bits
- (C) Byte
- (D) None of these

83. Decreasing the RAM causes _____

- (A) Fewer page faults
- (B) More page faults
- (C) Virtual memory get increases
- (D) Virtual memory get decreases

84. In the given language $L = \{ab, aa, baa\}$, __ number of strings are in L^*

- (1) baaaba
- (2) aabaaaa
- (3) baaabaaaabaa
- (4) baaabaaa

- (A) (1)
- (B) (2)
- (C) (3)
- (D) (4)

85. Consider the following functional dependencies in a database:

$A \rightarrow B$

$B \rightarrow C$

$D \rightarrow E$

$E \rightarrow D$

$F \rightarrow G$

$F \rightarrow H$

$(E, F) \rightarrow I$

The relation (E, D, A, B) is :

- (A) 2 NF
- (B) 3 NF
- (C) BCNF
- (D) None of the above